Roll No. Total No. of Pages : 02

Total No. of Questions : 13

B.Pharma (Sem.-6)

BIOPHARMACEUTICS AND PHARMACOKINETICS-THEORY

Subject Code : BP604T

M.Code : 77989

Date of Examination : 29-11-2023

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION-A

1. **Answer briefly :**

   a) Define the properties of drugs belonging to BCS Class I & II.

   b) Differentiate between relative and absolute bioavailability.

   c) Enumerate the parameters indicative of rate of drug absorption and elimination.

   d) Mention the advantages of IV administration of drugs.

   e) Mention two examples where particle size is known to influence drug absorption.

   f) Name the phases observed in a two compartment model.

   g) What are bioequivalent products?

   h) What is pinocytosis?

   i) What is renal clearance and how is it calculated?

   j) Why volume of distribution is called "apparent" and what is the unit of V_d?

SECTION-B

2. What is Non-Linear pharmacokinetics? Enumerate the reasons for non-linear pharmacokinetics and discuss its implications.

3. What is Sigma-Minus method? Explain the method of calculating elimination rate constant by this method with the help of suitable equations.

4. IV bolus dose (4 mg / kg) was administered to a patient of 75 kg weight. Following equation represented the drug kinetics: $C_p = 78\ e^{-0.46t}$

   Calculate: (a) $t_{1/2}$; (b) V_d; (c) Plasma concentration after 4 hr.

SECTION - C

5. Discuss the Complete Cross Over and Latin square designs with respect to bioequivalence trials.

6. Write a note on the approaches used for enhancing solubility and dissolution rates of poorly water soluble drugs.

7. Enumerate physiological factors influencing drug absorption and discuss any two in detail.

8. If the $t_{1/2}$ of a drug is 12 hr, how long would it take for 150 mg dose to get 30% eliminated?

9. Considering an IV bolus dose and simultaneous IV infusion, prove that for instantaneously achieving steady state, a loading dose of R/K (infusion rate/elimination half-life) is necessary.

10. Discuss the phase -I reactions for drug metabolism.

11. Comment on liver extraction ratio and explain its role in drug pharmacokinetics.

12. Write briefly about facilitated and active transport of drugs.

13. Comment on factors affecting renal clearance of drugs.

NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.

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